RUN-20261012T0226Z · ACCOUNTING AND REFERENCE CORRECTION

Degree-83 IC against the strong rho reference

Exact retained a=1 / K600 fixture · completed five-pair native diagnostic

One public target; five paired seed rounds and five same-seed A/A control pairs.

All 22 children completed and independently replayed; no process failures or caps.

Source baseline: a8194d616ed8c04489ec46177d369f1b1670360a.

> Observed IC/strong cost: 1.3823× prepared online and 218.0346× fresh-process wall from retained base. These are native paired geometric means on one Q and one host. The historical 8.1813× packed-reference observation does not establish a strong-reference advantage.

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SECTION 01

Diagnostic conclusion and the historical correction

> For this fixed point and frozen five-pair panel, the native IC/strong paired geometric mean is 1.3823 [1.0358, 1.8823] online and 218.0346 [172.3584, 285.5737] for fresh-process wall. Values above 1 mean greater IC cost. No controlled vs_rho row is promoted.

The source-bound comparison does not reproduce the previous synthesis’s interpretation of a degree-83 online lead against the required reference. Its 8.1813× median recorded rho/IC ratio used the old packed reference. That worker stores every step, scans the orbit in polynomial coordinates and inverts per step. Current reference rules exclude it as the primary strong control. The old scalar answers remain valid; its reference interpretation is corrected.

The fresh reference uses distinguished points, normal-basis canonicalization and batched inversion over 32 lockstep lanes in one CPU worker. Both fresh methods solve the exact same public Q without receiving its scalar. The standalone replay verifies every one of the 22 outcomes, including the two excluded warm-ups.

The intervals describe five seed/repeat pairs on this one Q and one host. They do not estimate performance on independently drawn targets, other hosts or deployed curves. The whole-process comparison begins with the retained base file and omits original base selection/construction; it is not a construction-from-curve total.

Recorded estimator	Value	95% native interval / n
IC/strong prepared-online cost	1.382330	[1.035774, 1.882300] · 5 pairs
IC/strong fresh-process wall	218.034569	[172.358384, 285.573727] · 5 pairs
Strong A / strong B online control	0.997036	[0.967928, 1.023496] · 5 pairs
Archived packed/IC median	8.181302	Historical only; different reference and estimator

Sources: R02, R03, R04, R07.

SECTION 02

Exact field, subgroup, point and frozen design

The correct exact-field slug is `icv1-f2m83-t6151469093347-9eba6cd5`. Full identity: `ICV1:f2m-83-a5fd0d39:6151469093347:967140655691088192855606 2:0x1:-7:unk:r:9eba6cd5d359`. The frozen protocol and original archive replay copied the wrong cdcc5432 alias, which belongs to low terms [0,1,2,45]. Their actual [0,2,4,7] field and G/Q arithmetic are unchanged. The separate erratum preserves frozen evidence.

Field degree 83 is distinct from the subgroup's approximately 52.928 bits. This is the retained a=1 instance, not the a=0 degree-83 / 81-bit ECC2K confidence gate. Exact polynomial, subgroup and G/Q coordinates are the primary binding.

The panel requests two excluded warm-ups, five same-seed strong/strong A/A pairs, and five interleaved strong/IC pairs; all 22 children completed verified. Strong seeds are 202610061–202610065 and IC rank seeds 21–25. Matched order alternates by round. One fixed Q is repeated in every child.

Item	Frozen value
Curve and field	$y^2+xy=x^3+x^2+1$; GF(2)[z]/(z^83+z^7+z^4+z²+1)
Prime subgroup r / cofactor h	8569786107849059 / 1128547018
Generator G	(661424465139818486029 2015,886826041002924939153 9529)
Public Q	(265589834553716618053 2960,388152630799167326529 2615)
Retained base	600 signed-Frobenius representatives; 99,600 usable points
IC workers	Frozen compact-orbit producer; rank threads 1, target threads 1
Strong policy	Wide-field exact-point adapter; 32 lockstep lanes; DP bits 4; step-cap factor 2000; fruitless escape false
Envelope	CPU 3, one pinned CPU; 12 GiB RLIMIT_AS; 1800 s process-group timeout

Sources: R01, R05, R06, R07.

SECTION 03

Three spaces and a one-dimensional subgroup

The arithmetic field has 83 coordinates over F2. The retained base is a point-defined union of 600 signed-Frobenius orbits, not a small linear subspace of those field coordinates. Its relation labels are formal vectors in (Z/rZ)^600. The cyclic prime-order subgroup itself has one scalar dimension over Z/rZ; 600 columns do not create 600 independent group directions.

A normal basis changes field representation and makes Frobenius squaring a rotation; it does not change factor-base membership. The displayed four-vector witness is retained, not newly discovered. Independent replay checks the actual four-point sum and scalar answer; the full 600-row preparation transcript remains missing.

One exact pipeline, three distinct spaces

Field degree: 83 · prime-subgroup size: approximately 53 bits · r = 8569786107849059

The cyclic subgroup has one scalar dimension over Z/rZ; the 600 IC coordinates are formal coefficient labels.

01 / FIELD COORDINATES

83 coordinates over F_2

$F = F_2[z] / (z^{83} + z^7 + z^4 + z^2 + 1)$

Polynomial and normal bases represent the same field values.

Squaring rotates normal coordinates. Least rotations define orbit keys.

Not a factor-base membership rule.

encodes →

02 / FACTOR-BASE POINT SET

600 orbit representatives

$B = \{ \varepsilon \tau^k(R_j) \}$

$0 \le k < 83$; $j = 0,\ldots,599$; $\varepsilon = \pm 1$

99,600 distinct signed-orbit points

Membership is defined by points. This B is not a linear subspace of the 83 field coordinates.

labels →

03 / RELATION COEFFICIENTS

Vectors in $(Z/rZ)^{600}$

$v(\varepsilon \tau^k(R_j)) = \varepsilon \lambda^k e_j$

e_j selects one orbit/log column. Coefficients track phase and sign.

These 600 column coordinates are separate from field encoding and the actual point-set B.

SIGNED-FROBENIUS TRANSPORT ON THE PRIME SUBGROUP

$\tau(x,y) = (x^2,y^2) \cdot -P = (x,x+y) \cdot \tau(P) = [\lambda]P$

$\lambda = 2998004823950457 \cdot \lambda^2 - \lambda + 2 = 0 \pmod r$ · exact order 83

RETAINED FOUR-POINT WITNESS: $P_1 + P_2 + P_3 + P_4 = Q$

Point indices [87298, 27061, 36601, 66281] · signs and shifts independently replayed in the archived receipt

$$v_Q = \lambda^{74} e_{525} - \lambda e_{163} - \lambda^{40} e_{220} - \lambda^{23} e_{399}$$

FACTOR-LOG DOT PRODUCT

$L_j = \log_G(R_j) \cdot d = v_Q \cdot L \pmod r$

→

INDEPENDENT FINAL REPLAY

$[d]G = Q$

Full factor-log/rank transcript is not retained: final witness replay does not certify every preparation equation.

`Exact legacy model: icv1-f2m83-t6151469093347-9eba6cd5 · primary binding: explicit field + r + G + Q`

Source: verification/INVARIANTS.md and verification/CURVE_ID_ERRATUM.md · correctness diagram; no runtime claim

Exact retained signed-orbit labels and four-vector witness, from INVARIANTS.md and the independent replay. The diagram describes existing algebra and supplies no runtime claim. [R04–R06]

Sources: R04, R05, R06.

SECTION 04

The invariant shared by IC and strong rho

On the subgroup, τ(P)=[λ]P with λ=2998004823950457, λ²−λ+2=0 mod r and exact order 83. Signed Frobenius multiplies each attached scalar coefficient by ελ^k. IC records that transport in its factor-log columns; rho records it in two state scalars for P=[a]G+[b]Q. Both are formal labels in the same one-dimensional subgroup.

At a rho distinguished-point collision, the nonzero denominator b1−b2 permits d=(a2−a1)/(b1−b2) mod r. A zero denominator yields no recovered scalar. IC instead extracts a four-point witness and applies its factor-log dot product. Each final answer requires [d]G=Q replay. The shared invariant does not make their preparation, target work or runtime equivalent.

The same invariant, two coefficient systems

83 F_2 field coordinates · one scalar dimension in the prime-order subgroup · λ = 2998004823950457

$C = \varepsilon\,\tau^k(P) = [\varepsilon\,\lambda^k]P$ · forward phase transport on the same exact group

IC / 600 FORMAL FACTOR-LOG COLUMNS

$v(P) \in (Z/rZ)^{600}$

$v(C) = \varepsilon\,\lambda^k\,v(P)$

For $P = R_j$: $v(P) = e_j$; $L_j = \log_G(R_j)$.

Root-index extraction yields a four-point witness and its coefficient vector v_Q.

$d = v_Q \cdot L \pmod r$

Full rank/log preparation transcript remains absent.

STRONG RHO / TWO FORMAL STATE SCALARS

$P = [a]G + [b]Q$

$(a,b) \to (\varepsilon\,\lambda^k a,\ \varepsilon\,\lambda^k b)$

Equal canonical collision points imply

$(a_1 - a_2) + (b_1 - b_2)d = 0 \pmod r$.

$d = (a_2 - a_1) / (b_1 - b_2) \pmod r$

Requires $b_1 - b_2 \neq 0$; zero gives no scalar.

Two labels do not mean a two-dimensional subgroup.

Independent final-answer replay: [d]G = Q

Same algebraic transport does not imply the same work or runtime. Source: verification/INVARIANTS.md.
Exact legacy model: icv1-f2m83-t6151469093347-9eba6cd5 · method comparison only; no fresh timing claim

Existing forward signed-Frobenius transport in the retained IC labeling and strong rho state. This is a method comparison, not a fresh identity or timing claim. [R05]

Sources: R05, R07.

SECTION 05

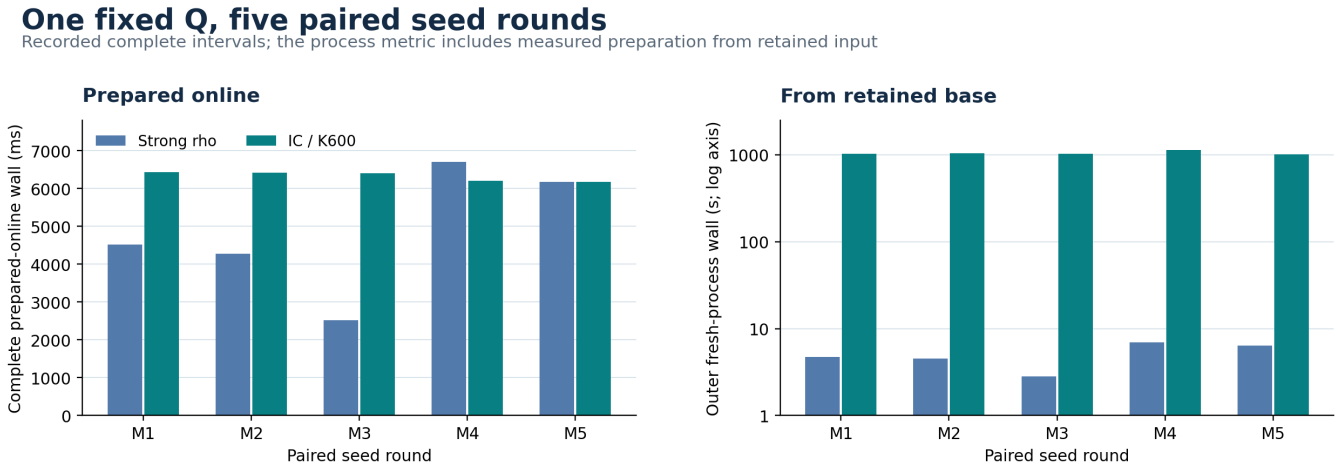
Per-round recorded online and process values

Online values are each producer’s complete declared target interval. The primary statistic does not replace those intervals with a sum of component clocks. Every matched row solves Q; there are no exclusions, failed answers, overall caps or missing matched receipts.

Strong online includes Q membership/subgroup validation, target-dependent starts, walk/collision work and generic scalar replay. The IC target stage assumes the retained fixture is prevalidated; that declared validation asymmetry remains visible.

M4 has a lower measured IC online interval: 6204.636907 ms versus strong 6698.575519 ms, native strong/IC ratio 1.079608. M5 is near a tie, native ratio 0.999600. The conditional paired geometric mean over all five rounds still gives greater IC cost; this does not assert that strong is faster in every seed round.

Pair / order	Strong online ms	IC online ms	Strong/IC native ratio
M1 · rho→IC	4516.159970	6437.368839	0.701554
M2 · IC→rho	4279.021477	6416.446390	0.666883
M3 · rho→IC	2514.613847	6408.147497	0.392409
M4 · IC→rho	6698.575519	6204.636907	1.079608
M5 · rho→IC	6176.824943	6179.295727	0.999600



Recorded intervals for run-20261012T0226Z, n=5 seed pairs on one Q. Right: fresh-process wall from retained base, logarithmic axis; includes measured reusable preparation. [R02,R03]

Sources: R02, R03, R04.

SECTION 06

Native uncertainty and same-seed A/A controls

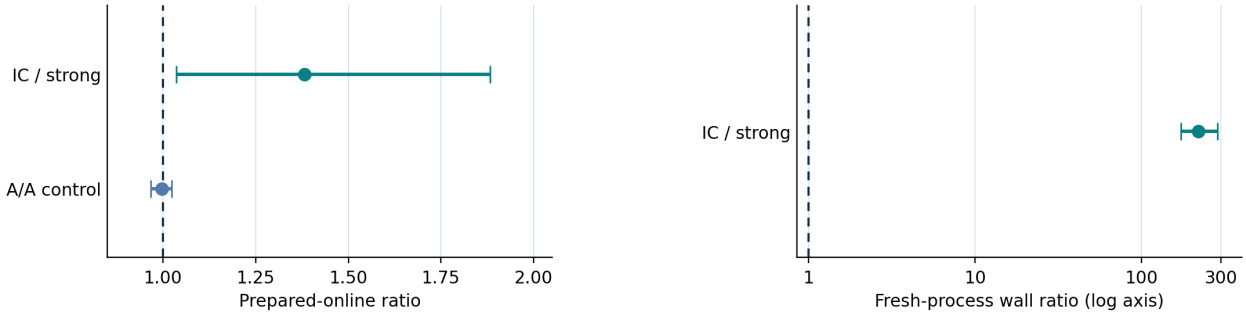
The frozen estimator takes the arithmetic mean of within-pair log(strong/IC) ratios and exponentiates it. It uses 20,000 paired percentile-bootstrap resamples. The runner applies seed 2026101283 to A/A and seed+1=2026101284 to matched pairs. The native secondary exporter presents IC/strong as the reciprocal, with interval [1/upper,1/lower]; it performs no new resampling or estimator change.

These geometric means are not ratios of arithmetic mean times. The A/A control’s paired geometric mean is 0.997036, interval [0.967928,1.023496], with observed online ratios spanning 0.948233–1.026892. It diagnoses repeat noise at fixed seeds on this host; it does not establish physical-host exclusivity.

n=5 is a small conditional panel. Seed sensitivity and repeated measurement of one Q cannot supply a fresh-target extraction-tail distribution. Keep all raw pairs, avoid population-wide significance claims, and prospectively freeze independent unseen targets for the next question.

Native paired geometric means and 95% intervals

IC/strong values are native reciprocals of the frozen strong/IC estimator; no new resampling



IC/strong > 1 means greater IC cost; parity is the dashed line. A/A compares two strong runs with the same seed.
Each interval is conditional on n=5 paired repeats of one Q and one host. It is not a fresh-target population interval.
Sources: statistics.json + native report_data_secondary.json · run-20261012T0226Z

Native paired geometric means and recorded 95% intervals, run-20261012T0226Z. IC/strong >1 means greater IC cost; A/A is strong-A/strong-B. Each interval covers five pairs on one Q and one host. [R02,R03]

Sources: R01, R02, R03, R10.

SECTION 07

Fresh-process cost from the retained base

The outer process starts from the retained factor-base file. IC loads its point arrays, builds the root index and acquires/solves its guided rank/log preparation. It does not newly select/build the original 600 representatives. OS page cache is not flushed. Thus cold means fresh process-local state, not OS-cold or total construction-from-curve runtime. Adding original construction cost would add IC work, but its unmeasured value stays unknown.

Strong curve/normal-basis construction and G-only jumps are measured. Supplied subgroup/catalog facts are parameters, not newly timed curve discovery or factorization. The native means below are secondary descriptive arithmetic means and ranges over five successful formal runs; they replace neither the paired primary estimator nor its interval.

Metric: mean [min,max]	Strong	IC
Preparation ms	191.516 [186.924, 194.744]	1047924.983 [1009528.197, 1139341.566]
Process wall s	5.068 [2.804, 6.909]	1054.552 [1015.960, 1145.812]
Child user+system CPU s	5.029 [2.709, 6.887]	1054.438 [1015.897, 1145.722]
Outer peak RSS bytes	116588544.0 [55263232.0, 208527360.0]	5811525222.4 [5811470336.0, 5811605504.0]

- IC prepared-online extraction is measured after its reusable rank/log preparation. Fresh cold children reconstruct index/log state independently; no cross-process warm cache is inherited.
- The retained root table has 29,878,182 entries. Table-entry counts are stored states, not independently verified factor logs or useful-row counts.
- The root/index and guided preparation must stay charged in this workload. Amortized reuse across new targets requires an explicitly different workload count.

Sources: R03, R05, R07.

SECTION 08

Phase residuals and the outer wall boundary

IC PDP subtracts its nested final-lift check from extraction wall at one target thread. That check is only the final addition/equality; earlier lift work stays under PDP. All five formal IC rows have exact five-phase conformance=false. Strong's three component timers also leave nonzero residuals; its five-phase flag is inapplicable/null. The native primary uses complete online intervals.

Outer wall starts before spawn and ends when wait4(WNOHANG) observes exit. The runner polls every 100 ms: wall includes nominal 0–100 ms detection delay plus scheduling/poll/heartbeat effects, not a strict 100 ms bound. Child CPU excludes parent sleeping. Formal M3 strong reports producer process_before_receipt_io=2.707912963 s, CPU=2.708705 s and parent wall=2.803706954 s. These boundaries do not prove an exact causal overhead decomposition. Frozen wall values remain untouched.

IC retains five component clocks. Each residual below is the native complete interval minus its reported phase sum; no phase sum is silently substituted for online.

Pair	Query ms	PDP ms	Lift ms	Descent ms	Replay ms	Residual ms
M1	0.010927	6437.211479	0.012360	0.000611	0.103506	0.029956
M2	0.000030	6416.329133	0.012258	0.000640	0.103356	0.000973
M3	0.000020	6408.030341	0.012148	0.000601	0.103426	0.000961
M4	0.000020	6204.505849	0.025700	0.000630	0.103586	0.001122
M5	0.000071	6179.177809	0.012648	0.000831	0.103577	0.000791

Strong reports Q query/validation, walk/collision and generic replay. The different component definitions are preserved.

Pair	Q validation ms	Walk/collision ms	Generic replay ms	Residual ms
M1	0.232440	4515.698654	0.228475	0.000401
M2	0.232550	4278.561313	0.227333	0.000281
M3	0.248956	2514.137047	0.227503	0.000341
M4	0.261765	6698.084148	0.229306	0.000300
M5	0.233803	6176.356336	0.234494	0.000310

Both native producers have nonzero timer residuals. The complete-interval primary and descriptive process metrics remain available as diagnostic observations; exact phase-accounting admission remains unmet.

The parent observer-clock caveat is retained: before-spawn→observed-exit includes polling and scheduling/heartbeat effects. CPU and producer process timestamps corroborate it without retroactively adjusting wall. Future harnesses should use event-driven pidfd/SIGCHLD or native blocking wait timing.

Sources: R03, R07, R10.

SECTION 09

Counters and independent correctness at their actual scope

> Independent answer/base/witness correctness is established at the stated replay scope. It does not certify every missing preparation equation, reproducibility of wall time, physical-host isolation or a speedup across independent targets.

IC returns the same four indices [87298,27061,36601,66281] and 8,845,441 target probes in each formal repeat of Q. Probe units count partner candidates reaching canonicalization/lookup, not every S3 call or group addition. The producer records rank=600, 600 admitted rank relations and zero rank failures; the complete rank equations and factor-log vector are not exported.

The standalone C++ replay revalidates the field/curve/subgroup, all 600 representatives and 99,600 point/label records, every actual IC four-point witness and all 22 recovered answers. It reports PASS, 22 independently verified answers and zero invalid rows. The validation-only scalar sidecar is checked after answers, not passed to either solver.

Rho steps and IC probes are different counted units. The strong lane counters below describe internal lane retirement and cycle handling; nonzero local capped_walks do not mean a process failed or reached the overall cap. Common calibrated operation-equivalent S stays null.

Pair	IC probes	Rho walk steps	Local capped lanes	Fruitless cycles
M1	8845441	10203425	148	1845
M2	8845441	9436758	147	1699
M3	8845441	5353957	87	973
M4	8845441	14229982	234	2510
M5	8845441	13159092	187	2319

Sources: R03, R04, R05, R07.

SECTION 10

Host, validation and admission boundaries

The host receipt describes virtualized KVM Linux x86_64, kernel 6.18.44, AMD EPYC 9V74 exposed model, five guest logical cores and one NUMA node. Cgroup RAM is 16 GiB with CPU quota 4. Each child has CPU-3 affinity and a 12 GiB virtual-address cap. Native isolation finishes exit 0 with contended_samples=0 and threads_moved=29. These are guest monitoring observations, not physical-host exclusivity or sealed ecbench L2.

All 22 children completed verified: two excluded warm-ups, ten A/A children and ten matched children. There are no process failures, overall caps or excluded pairs. Fresh independent arithmetic replay verifies all 22 answers. Final default-feature cargo test --release --locked --lib passes 3947 tests with zero failures and 112 ignored (260.41 s). Delivery runner tests pass 7/7, native exporter 4/4, and final combined example tests pass IC 10/10 plus strong 2/2. These validation checks are separate from timed execution.

The frozen baseline IC executable predates the isolated discarded-RNG-draw collector repair. The strong adapter was compiled after it, but neither measured solver calls that generic collector path. The repair is a reproducibility fix outside this measured algorithm, not an IC optimization or a reason to rewrite frozen timing receipts.

The exact legacy wide-field IC adapter is absent from the current sealed ecbench path. Canonical candidate/workload identities remain null, common S is uncalibrated, the full rank/log transcript is missing and there is no second-host timing/replay admission. Conductor invocation returned command-not-found, and no tracked .conductor/WORKFLOW.md was available. These gaps remain explicit; this report does not promote vs_rho.

- Correct the old packed-reference interpretation and wrong-modulus alias in new prose; preserve old frozen receipts.
- Next measure independent unseen targets with a prospectively frozen preparation/selector policy; five seed repeats of Q do not resolve target tails.
- For full algebraic preparation certification, retain every admitted rank relation, its witness and setup scalar, rank/matrix evidence, the solved factor logs and independent [L_j]G=R_j replays.
- For a construction-from-curve objective, newly measure original representative selection/construction; for admission, add exact ecbench integration, calibrated common costs and applicable isolation/second-host gates.
- Use event-driven child completion in a future runner, while retaining CPU totals and producer timestamps; leave this frozen panel’s wall unchanged.

Sources: R04, R07, R08, R09, R10. Validation logs: validation_logs/full_lib_tests_final.log and validation_logs/examples_tests_final.log.

SECTION 11

Evidence register and reproducible report source

This is an accounting/reference correction and a source-bound diagnostic of the existing retained candidate. It introduces no selector, algebraic identity, asymptotic improvement or admitted performance row. The quantitative plots lay out native recorded values and intervals; Python performs no statistical estimation, arithmetic replay or benchmark computation. The authoritative executed identity is records/frozen_panel_inputs.sha256, checked unchanged after the panel and corroborated by every child start/execution receipt. Earlier frozen_before_execution hashes are preparation history only and are not used as executed bindings.

The report lives beside the raw records at research/n83_evaluation_20261012/reports/run-20261012T0226Z. Native JSON copies under data/ preserve the recorded source outputs; figures.py, editable SVG diagrams, report_content.json and report_source.md reproduce its layout.

Canonical graph review found the n83 retained scoreboard anchor concerns the distinct a=0 / 81-bit engineering capacity study. The boundary ledger already labels the old a=1 observation exploratory; no controlled frontier or ledger row is promoted here. The scope of graph/source checks is recorded in the study README.

Evidence / source pin	SHA-256 or exact reference
Native primary statistics	43b00e963b379aaf718a16b864a3125a866a8be9f7f83b8eaa109d23c3d1560e
Native secondary export	b8101e76589558c8329831e3fe4649ea010661a34c6653bdc2fec40b5063b0b3
Independent fresh replay	980a3d7d52db96a559dd36ad12e317c9b12be9f1c161a2d0d86803aafcc72441
Paired-record input	e1f1b019ed5f80a739b0acc9c2e7f971d8e3400f113b47700067b0a64f8c0215
Frozen IC executable	983f28400b7852fc6628ba97d3ea83b2ab9e32d9944658c7eea831ee89a4abf2
Frozen strong executable	98949b7127baeb1fa55420d23dc13604069fcc8fde8d92b8f2e6bebd09718542
Executed native runner	8da1b77f9521fccdc5d8490a493146f0a6e2af92a2fab07c4e2e374954b9d2b4
Executed input protocol	ac7266f6caad2a4142ae06884ee748f980f9ee200c4e2649a0149f59e91adf29

Native source hashes above identify the copied recorded outputs and executables. Links below lead to the durable study records and source explanations.

ID	Evidence / path
R01	Frozen protocol: runner/protocol.json
R02	Native paired statistics: records/panel/statistics.json
R03	Native secondary export: records/panel/report_data_secondary.json
R04	Independent fresh replay: verification/fresh_replay_receipt.json
R05	Three spaces and signed-orbit witness: verification/INVARIANTS.md
R06	Curve-identity erratum: verification/CURVE_ID_ERRATUM.md
R07	Source audit: source_audit/AUDIT.md
R08	Host and isolation records: records/README.md
R09	Study delivery and validation: README.md
R10	Native runner source: runner/src/main.rs
R11	Native secondary exporter: descriptive_summary/src/main.rs

Sources: R01, R02, R03, R04, R05, R06, R07, R08, R09, R10, R11.